LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q1 · Rank 1: Foundation

Calculate $(-2)(-3)$.

A -6

B 6

C -5

D 5

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q2 · Rank 1: Foundation

Calculate $-2+(-3)$.

A 1

B 5

C -5

D 6

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q3 · Rank 1: Foundation

In $x^2 - x - 6$, what is the coefficient b in $x^2 + bx + c$?

A -1

B 1

C -6

D 6

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q4 · Rank 1: Foundation

Expand $x(x + 5)$.

A $x^2 + 5$

B $2x + 5$

C $6x^2$

D $x^2 + 5x$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q5 · Rank 1: Foundation

For the numbers 3 and 4, which ordered pair gives (product, sum)?

A $(7, 12)$

B $(12, 7)$

C $(12, 1)$

D $(7, 1)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q6 · Rank 1: Foundation

Which pair has product 6 and sum 5?

A 1 and 6

B −2 and −3

C 2 and 3

D −1 and −6

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q7 · Rank 1: Foundation

Which pair has product 6 and sum −5?

A −2 and −3

B 2 and 3

C −1 and −6

D 2 and −3

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q8 · Rank 1: Foundation

Expand $(x-2)(x-3)$.

A x^2+5x+6

B x^2-x-6

C x^2-5x-6

D x^2-5x+6

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q9 · Rank 2: Core

Factor $x^2 + 5x + 6$.

A $(x + 2)(x + 3)$

B $(x - 2)(x - 3)$

C $(x + 1)(x + 6)$

D $(x - 1)(x - 6)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q10 · Rank 2: Core

Factor $x^2 + 7x + 12$.

A $(x + 1)(x + 12)$

B $(x + 2)(x + 6)$

C $(x + 3)(x + 4)$

D $(x - 3)(x - 4)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q11 · Rank 2: Core

Factor $x^2 - 5x + 6$.

A $(x + 2)(x + 3)$

B $(x - 2)(x - 3)$

C $(x - 1)(x - 6)$

D $(x - 2)(x + 3)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q12 · Rank 2: Core

Factor $x^2 + x - 6$.

A $(x - 3)(x + 2)$

B $(x + 3)(x + 2)$

C $(x + 6)(x - 1)$

D $(x + 3)(x - 2)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q13 · Rank 2: Core

Factor $x^2 - x - 6$.

A $(x+2)(x-3)$

B $(x+3)(x-2)$

C $(x-2)(x-3)$

D $(x-6)(x+1)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q14 · Rank 2: Core

Factor $x^2 + 6x + 9$.

A $(x+9)(x+1)$

B $(x-3)^2$

C $(x+3)^2$

D $(x-3)(x+3)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q15 · Rank 2: Core

Factor $x^2 - 9$.

A $(x-3)^2$

B $(x-3)(x+3)$

C $(x+3)^2$

D $(x-9)(x+1)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q16 · Rank 2: Core

Factor $x^2 + 5x$.

A $x(x-5)$

B $(x+5)^2$

C $(x+1)(x+5)$

D $x(x+5)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q17 · Rank 3: Application

Solve $x^2 + 5x + 6 = 0$.

A $x = 2$ or 3

B $x = -3$ or -2

C $x = -6$ or -1

D $x = -3$ or 2

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q18 · Rank 3: Application

Solve $x^2 - 5x + 6 = 0$.

A $x = -3$ or -2

B $x = -3$ or 2

C $x = -2$ or 3

D $x = 2$ or 3

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q19 · Rank 3: Application

Solve $x^2 + x - 6 = 0$.

A $x = -3$ or 2

B $x = -2$ or 3

C $x = -3$ or -2

D $x = 2$ or 3

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q20 · Rank 3: Application

Solve $x^2 + 5x = 0$. Include all real solutions.

A $x = -5$ only

B $x = 0$ only

C $x = -5$ or 0

D $x = 0$ or 5

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q21 · Rank 3: Application

A proposed factorisation of $x^2 - 5x + 6$ is $(x - 1)(x - 6)$. What middle coefficient does the proposed product actually produce?

A -5

B -7

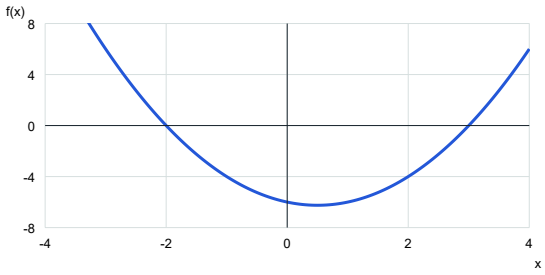
C 5

D 7

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q22 · Rank 3: Application

For $f(x) = x^2 - x - 6$, which two points are the horizontal-axis intercepts?



The graph of f(x)=x²−x−6. Locate the two crossings of y=0.

A $(-3, 0)$ and $(2, 0)$

B $(-3, 0)$ and $(-2, 0)$

C $(2, 0)$ and $(3, 0)$

D $(-2, 0)$ and $(3, 0)$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

LECTURE 3 SUPPLEMENT

Factoring Quadratics: Step by Step

24 single-choice questions. Ranks 1–3 build the core skills. Rank 4 is optional. Work at your own pace.

Q23 · Rank 4: Challenge (optional)

Suppose $x^2 + kx + 12 = (x + m)(x + n)$, where m and n are positive integers. What is the smallest possible k?

A 5

B 6

C 7

D 12

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.

Q24 · Rank 4: Challenge (optional)

Which expression cannot be factored as $(x + m)(x + n)$ with integer m and n?

A $x^2 + x - 6$

B $x^2 - 5x + 6$

C $x^2 + 6x + 9$

D $x^2 + x - 1$

Original supplementary questions based on Jacques §2.1, printed pp.127–128 / PDF pp.144–145.